Electricity Scheduling for a Flexible Data Centre Workload

A retrospective study of electricity cost and carbon intensity in Great Britain

Independent Research Project

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Abstract

This project asks a practical question: can forecasts of electricity prices help schedule a flexible data centre workload at lower cost? It uses half-hourly electricity data from Great Britain for 2024–2025. Two forecasting models, Ridge regression and Gradient Boosting, are compared with a seven-day seasonal baseline. Each forecast is passed to a linear optimisation model that schedules a hypothetical 6 MWh daily workload within a 1 MW power limit. The resulting schedule is evaluated using the electricity prices and national average carbon intensity observed afterwards.

On a corrected retrospective test covering 183 complete days in July–December 2025, Gradient Boosting reduced modelled electricity cost by **20.46%** compared with uniform scheduling. The corresponding reduction in attributed emissions was **18.02%**. Ridge had lower forecasting error, but Gradient Boosting produced better daily price rankings and lower realised scheduling cost. The results show why a forecasting model should be evaluated against the decisions it supports. They apply to a simplified, fully flexible workload and a wholesale-indexed cost model; they do not represent verified savings from an operating data centre.

1 Research question and motivation

Electricity prices change throughout the day. Some data centre tasks, such as selected batch calculations or data processing jobs, may be moved to a different time if they have enough time before completion is required. This creates a scheduling opportunity: process the same amount of work during lower-price periods without exceeding available computing capacity.

The problem is that tomorrow's prices are not known when the schedule must be chosen. A forecast is needed, and forecast errors may lead to poor decisions. A model with low average prediction error may still fail to identify the cheapest periods. For a scheduling application, that distinction matters.

The main research question is:

> *How much can forecast-informed scheduling reduce the modelled electricity cost of a flexible data centre workload, and how reliable are those savings outside the model-development period?*

The study also examines two related questions. First, do changes in electricity cost coincide with changes in estimated emissions based on national average grid carbon intensity? Second, how sensitive are the savings to power capacity and workload deadlines?

2 Data and preparation

2.1 Data sources

The data cover 1 January 2024 to 31 December 2025 at 30-minute resolution. This gives 35,088 timestamps on a continuous UTC time grid, including the leap year 2024. The study uses the following sources:

- **Electricity prices:** Elexon Balancing Mechanism Reporting Service market index data, using the APX market index data provider (`APXMIDP`). Prices are in £/MWh, and the dataset includes a market volume field. Market index prices describe wholesale short-term trading; they are not retail tariffs charged to data centres [1].
- **Carbon intensity:** Great Britain's national Carbon Intensity API, associated with the National Energy System Operator (NESO). The 'actual' series is expressed in gCO_2/kWh and represents an estimate based on generation data [2, 3].
- **Electricity demand and generation:** Public half-hourly series used during exploratory data analysis to examine electricity-system behaviour and data quality. They are *not* included as explanatory variables in the final forecasting models.

The distinction between the price measure and a retail electricity bill is important. The simulated cost is obtained by multiplying the market index price by scheduled energy consumption. It excludes transmission and distribution charges, supplier fees, demand charges and taxes. Real facilities may also have fixed-price contracts or other tariff structures.

2.2 Cleaning and alignment

All timestamps were converted to UTC and aligned to the same 30-minute grid. Duplicate timestamps and unexpected gaps were checked. The final forecasting pipeline retains missing observations instead of inventing market prices.

One APX observation was unavailable in the source data. Another 11 records had zero market volume; these were treated as unavailable prices for model evaluation. Some carbon intensity observations were also missing. Missing actual outcomes were not filled to create artificial ground truth.

Earlier exploration identified periods with unusual zero reported fuel generation and differences in the coverage of some interconnector fields. These issues affected exploratory interpretation, but the final price and carbon forecasts do not use demand or generation values as inputs. This avoids adding uncertain or unavailable system variables to the forecasting model.

3 Forecasting design

3.1 The information available when a decision is made

The schedule for a target day is assumed to be decided at **00:00 UTC on the previous day**. For example, the schedule for 1 July is selected at 00:00 UTC on 30 June. A conservative **24-hour observation delay** is imposed on the available historical data. This delay is a modelling assumption; the actual publication time of each historical record has not been verified.

The target day contains 48 half-hour periods. For a target timestamp t, the model uses prices from 144, 192, 336 and 384 half-hour periods earlier, corresponding to three, four, seven and eight

days. These lags place the inputs before the assumed information cutoff for every target period in the day.

The model also uses calendar information known in advance: time of day, day of week and day of year. Each calendar variable is represented by sine and cosine terms. For a recurring quantity with period T, these take the form

$$\sin\left(\frac{2\pi z}{T}\right), \qquad \cos\left(\frac{2\pi z}{T}\right). \tag{1}$$

This representation lets the model learn recurring patterns without treating the end and beginning of a cycle as unrelated. For example, 23:30 and 00:00 are close in time even though their numerical hour values are far apart.

3.2 Models and baselines

Three methods are evaluated:

- **Seven-day seasonal naive:** Predict the price using the observation from the same half-hour seven days earlier. This is a simple benchmark that uses the weekly cycle.
- **Ridge regression:** Fit a linear combination of lagged prices and calendar features, with a penalty on large coefficients. Features are standardised inside the training pipeline. The selected regularisation parameter is $\alpha = 30{,}000$ [4].
- **Histogram Gradient Boosting:** Combine a sequence of decision trees to capture nonlinear patterns. The selected settings are 200 boosting iterations, learning rate 0.05, at most 15 leaves per tree and L2 regularisation 10 [5].

The models were selected using a validation period. Their settings were then held fixed for the final evaluation. Ridge and Gradient Boosting forecast the same price series and use the same input features.

3.3 Chronological evaluation

Randomly shuffling observations would mix early and late periods, which would be inappropriate for forecasting. The study therefore follows chronological divisions:

- **Development training:** historical observations before 30 March 2025.
- **Validation:** 1 April–30 June 2025, used to compare models, examine scheduling policies and select settings.
- **Retrospective test:** 1 July–31 December 2025, used to evaluate the selected approach on later data.

For the corrected final test, the price models were fitted once using labels **strictly before 29 June 2025 at 00:00 UTC**. This prevents the training labels from using records beyond the assumed data cutoff at the first test forecast origin. The model is then kept fixed during the test; historical lag features change as time passes.

If a model forecast is unavailable because a historical input is missing, the pipeline falls back to the same half-hour's observed price seven days earlier, then 14 days earlier, then 21 days earlier. This rule is based only on historical observations. It was added after an initial examination of test coverage and is disclosed as a correction to the retrospective evaluation. The final results must therefore not be described as those of a completely untouched test.

3.4 Forecasting metrics

Forecast quality is assessed using mean absolute error (MAE), root mean squared error (RMSE) and the average within-day Spearman rank correlation. For n actual prices y_i and forecasts $\hat{y}_i$:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{2}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}. \tag{3}$$

MAE measures the typical size of a prediction error. RMSE gives extra weight to large errors. Spearman correlation measures how well the model ranks the half-hours from lower to higher prices each day. Ranking is useful because the scheduling model must identify the cheapest periods.

4 Workload scheduling model

4.1 Operating assumptions

The optimisation problem represents a hypothetical collection of flexible computing tasks. Every day, the workload must use **6 MWh of electricity**. The available processing power is at most **1 MW**. Each half-hour can therefore contain at most 0.5 MWh of scheduled energy. All work is available from the beginning of the target day and must be completed within 24 hours.

Let x_t be the scheduled energy in MWh during half-hour t, where $t = 1, \ldots, 48$. Let $\hat{p}_t$ be the forecast electricity price in £/MWh. The objective is

$$\min_{x_1,\ldots,x_{48}} \quad \sum_{t=1}^{48}\hat{p}_t x_t, \tag{4}$$

$$\text{subject to} \quad \sum_{t=1}^{48} x_t = 6, \tag{5}$$

$$0 \le x_t \le 0.5 \qquad (t = 1, \ldots, 48). \tag{6}$$

Equation (5) ensures that the same amount of work is completed. Equation (6) enforces the maximum power of 1 MW. The model uses continuous x_t, so a workload can be divided across time periods. It does not model individual job arrivals, task dependencies or switching costs.

This is a **linear programme (LP)** because both the objective and constraints are linear. It is solved using SciPy's `linprog` function with the HiGHS solver [6]. With a linear cost objective and a capacity limit, the solution generally allocates as much work as possible to the periods forecast to be cheapest, subject to completing the required energy.

4.2 Comparison schedules

Every policy uses exactly 6 MWh and is evaluated on the same days:

- **Uniform:** use 0.125 MWh in each of the 48 periods. This is the main baseline and corresponds to constant power of 0.25 MW.
- **Fixed:** use 0.5 MWh in each of the first 12 periods (00:00–06:00 UTC).

- **Seasonal naive, Ridge, Gradient Boosting:** choose the LP schedule using the price forecast from the named model.
- **Oracle:** optimise using the actual future prices. This is a perfect-information benchmark, not a deployable forecast.

Once a schedule is chosen, the realised electricity cost is calculated using observed prices p_t:

$$\text{Cost} = \sum_{t=1}^{48} p_t x_t. \tag{7}$$

The model therefore makes decisions using forecasts and judges the results against later observations. This separation is necessary to assess how forecasting errors affect decisions.

4.3 Estimating associated emissions

If c_t is observed national average carbon intensity in gCO_2/kWh, the project's attributed emissions measure is

$$\text{Attributed emissions (kg } CO_2) = \sum_{t=1}^{48} c_t x_t. \tag{8}$$

The units work because 1 MWh is 1,000 kWh and 1 kg is 1,000 g. This measure estimates the emissions associated with the timing of electricity use at the national average intensity. It is **not** a causal estimate of emissions avoided on the electricity grid. A change in one facility's demand could affect marginal generation differently.

5 Results

5.1 Test coverage and forecasting performance

The retrospective test has 184 calendar days. Of these, **183 days** contain all 48 half-hours and the required actual outcomes and forecasts. One day, 6 September 2025, is excluded because an actual price observation is unavailable. The remaining 183 days give **8,784 evaluated half-hours**.

Historical fallback supplies 11 evaluated predictions for the seven-day naive baseline and 41 predictions each for Ridge and Gradient Boosting. The evaluation includes the fallback forecasts in the reported accuracy and cost metrics.

Table 1: Forecast accuracy on 8,784 test half-hours. Lower MAE and RMSE are better; higher daily Spearman correlation is better.

Model	MAE (£/MWh)	RMSE (£/MWh)	Mean daily Spearman
Seven-day naive	24.149	35.274	0.553
Ridge	**17.548**	**26.168**	0.626
Gradient Boosting	19.386	26.941	**0.677**

Ridge has the smallest errors in predicting price levels. Gradient Boosting ranks prices within each day more accurately. This gives a reason to examine the schedules instead of choosing a model only from MAE or RMSE.

5.2 Realised scheduling costs and attributed emissions

Table 2: Corrected retrospective test results for 183 days, relative to uniform scheduling. All policies complete the same 6 MWh daily workload.

Policy	Cost (£)	Cost saved (%)	Carbon (kg)	Carbon lower (%)
Uniform	80,769.28	0.00	137,597.75	0.00
Fixed	69,032.79	14.53	124,670.00	9.40
Seven-day naive	67,975.76	15.84	116,858.50	15.07
Ridge	66,054.29	18.22	116,840.50	15.09
Gradient Boosting	**64,247.33**	**20.46**	**112,798.00**	**18.02**
Oracle	57,092.73	29.31	110,814.50	19.47

Savings are calculated from the sum of costs on the same evaluation days:

$$\text{Cost saving (\%)} = 100\left(1 - \frac{\text{Cost}_{\text{policy}}}{\text{Cost}_{\text{uniform}}}\right). \tag{9}$$

Gradient Boosting reduces total simulated cost by **£16,521.95** compared with uniform scheduling, a reduction of **20.46%**. It produces a lower cost than Ridge and the seven-day baseline. The attributed emissions difference from uniform scheduling is **24,799.75 kg CO_2**, or **18.02%**. The main optimisation objective is cost; the emissions difference is measured afterwards.

The oracle cost is lower because it has perfect knowledge of prices. Its result represents the best achievable cost within this simplified feasible set using actual prices. The difference between the oracle and forecast-informed schedule shows that forecast errors still matter.

5.3 Uncertainty in the test savings

The project uses a paired, circular moving-block bootstrap with seven-day blocks and 5,000 samples. Each sample resamples groups of consecutive evaluation days and preserves the pairing between policies. Savings are recomputed using total costs within each sample. This is preferable to averaging daily percentage savings, which can become extreme when a day's baseline cost is small.

Table 3: Observed performance of Gradient Boosting and approximate 95% seven-day block-bootstrap intervals.

Comparison	Improvement (%)	Approx. 95% interval
Cost vs uniform	20.46	[16.66, 24.91]
Cost vs Ridge	2.74	[0.53, 5.42]
Cost vs seven-day naive	5.48	[2.92, 8.55]
Attributed emissions vs uniform	18.02	[15.04, 21.17]

The intervals suggest that the cost benefit is not driven entirely by a small number of unusual days. The intervals for the cost comparisons with Ridge and the seven-day baseline are above zero. These are approximate uncertainty estimates. Seven-day blocks do not capture every form of dependence, and the intervals do not include all uncertainty from model selection or retrospective pipeline correction.

6 Sensitivity analyses on validation data

The following experiments were conducted on **87 complete validation days**. They are used to understand the model's behaviour; they are not additional untouched test results.

6.1 Processing capacity

Daily energy remains fixed at 6 MWh, while the maximum available power is changed. Each scenario is compared with the same uniform 0.25 MW schedule.

Table 4: Cost savings from forecast-informed scheduling as maximum power changes.

Maximum power (MW)	Savings vs uniform (%)
0.25	0.00
0.50	15.57
0.75	19.40
1.00	22.51
1.50	25.28

At 0.25 MW, all 24 hours are required to complete 6 MWh, so the schedule cannot move energy between periods. Higher capacity gives the optimiser more choices. The observed savings increase as capacity rises, although a real facility would need to account for the cost and availability of extra capacity.

6.2 Completion deadlines

The next experiment requires all 6 MWh to be completed within 12, 16, 20 or 24 hours after midnight. Maximum power remains 1 MW. Each deadline is compared with a uniform schedule that finishes by the same deadline.

Table 5: Effect of the allowed completion window. Baselines are recalculated for each deadline.

Completion deadline	Savings vs corresponding uniform schedule (%)
12 hours	8.67
16 hours	14.43
20 hours	21.16
24 hours	22.51

Short deadlines restrict the periods available to the optimiser. A full-day window provides more opportunities to move work to lower forecast prices. Since the uniform baseline also changes with the deadline, the percentage savings should be interpreted as a within-scenario comparison.

6.3 Carbon-aware scheduling

A separate validation experiment forecast carbon intensity using a Ridge model with past carbon intensity and calendar features. It then minimised predicted price subject to a predicted

carbon budget. The budget was varied between the emissions of a cost-optimal schedule and a carbon-optimal schedule, both measured using the carbon forecast.

Intermediate carbon constraints reduced observed attributed emissions modestly relative to cost-only scheduling, but they increased observed costs. The carbon-only schedule did not give the lowest observed emissions. Across the 87 validation days, its emissions were 266 kg higher than those of cost-only scheduling, despite being lower under the carbon forecast. Paired bootstrap intervals for the intermediate changes in emissions included zero.

This result matters because an optimisation model can satisfy its constraints for *predicted* carbon intensity while performing differently under the *observed* values. A carbon-focused operational claim would require stronger carbon forecasts and further validation. The final primary policy therefore remains cost-only scheduling.

7 Discussion and limitations

7.1 Why forecast ranking matters

Ridge has the smallest test MAE and RMSE, but Gradient Boosting has the highest within-day ranking correlation and produces the lowest simulated cost among the practical forecast-based policies. The scheduling objective depends on selecting low-price periods. A small prediction error is useful, but it does not guarantee the best ordering of periods.

This finding supports evaluating forecasting methods using the application they serve. For this project, the realised scheduling cost is the main measure of decision quality. Forecast accuracy remains important because it helps explain the result and diagnose poor predictions.

7.2 Limits of the retrospective experiment

There are several reasons why the numerical savings should not be presented as savings from a deployed data centre:

- **Price measure:** APX market index prices are wholesale-market observations. Real electricity bills may contain fixed and variable charges that limit the value of shifting demand.
- **Workload model:** All 6 MWh are assumed to be flexible, divisible and ready at midnight. Real workloads have release times, job deadlines, service requirements, task dependencies and potentially a non-flexible base load.
- **Forecast timing:** The 24-hour data delay is assumed. Original real-time publication timestamps and archived forecast vintages have not been checked. The work is a reconstructed historical experiment, not a verified live forecasting system.
- **Missing data and test correction:** One test day lacks a complete actual price outcome. The historical fallback rule was added after the first test coverage inspection, and the training cutoff was subsequently corrected. The final reported test is therefore a *corrected retrospective evaluation.*
- **Emissions interpretation:** National average carbon intensity does not directly measure the marginal emissions response to moving electricity demand. The reported 18.02% is a change in attributed emissions under this accounting rule, not verified avoided emissions.
- **Statistical uncertainty:** The seven-day bootstrap is approximate and does not cover all market regime changes, modelling choices or longer forms of time dependence.

7.3 Possible extensions

A practical extension would use archived point-in-time market data to establish exactly what information was available at each decision time. The workload model could then include individual job arrival times, hard deadlines, a fixed base load and realistic tariffs. A rolling backtest could retrain forecasts at specified times using only data released by that time. For carbon-aware scheduling, marginal emissions estimates or a more carefully validated carbon forecast would provide a stronger basis for interpretation.

8 Conclusion

This project combined historical electricity price forecasting with linear optimisation of a hypothetical flexible data centre workload in Great Britain. The corrected retrospective evaluation covers 183 complete days in July–December 2025. Gradient Boosting scheduling reduced modelled wholesale-indexed electricity cost by **20.46%** relative to uniform scheduling, with an approximate 95% block-bootstrap interval of **16.66%–24.91%**. Associated emissions calculated from national average carbon intensity were **18.02%** lower.

Ridge produced more accurate price-level forecasts, while Gradient Boosting ranked intraday prices more effectively and produced better scheduling decisions. The main lesson is that forecasting performance should be judged in the context of the operational decision. The reported savings are promising for the simplified model. Confirming real-world savings would require a more detailed workload model, realistic electricity tariffs and verified point-in-time data availability.

References and documentation

References

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